

Information Security Project Proposal

Project Title:

Secure File Distribution System with Time-Limited Access Control

Student Information:

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1. Problem Statement / Project Idea

In today's digital environment, files are frequently shared through emails, messaging applications, and cloud platforms. However, once a file is shared, the sender loses control over how the file is accessed, copied, or distributed. There is no built-in mechanism to restrict unauthorized access or limit how long a file remains accessible.

This lack of control can lead to serious security issues, especially when sensitive or confidential data is involved. Files may be accessed by unintended users or remain available indefinitely without any restrictions.

This project aims to address these issues by developing a Secure File Distribution System with Time-Limited Access Control. The system will ensure that files are encrypted before sharing, accessible only through password authentication, and restricted by an expiry time. After the expiry time, access to the file will be denied automatically.

The main objective is to demonstrate key information security principles such as confidentiality, access control, and data protection using a practical implementation.

2. Scope

The project will focus on developing a desktop-based application that provides secure file sharing with controlled access. The system will include the following features:

- File encryption before sharing
- Password-based authentication for file access
- Time-based expiry mechanism to restrict access duration

- Decryption of files only after successful verification
- Basic logging of access attempts

The system will operate locally and will not include network or cloud integration. The encryption method used will be a simplified approach suitable for learning purposes rather than enterprise-level security.

Despite these limitations, the project will effectively demonstrate how encryption and access control mechanisms can be combined to enhance the security of file sharing.